Stage 8 Science — Practice Test 1

Paper 1: 45 minutes · calculator allowed · Paper 2: 45 minutes · Total 100 marks

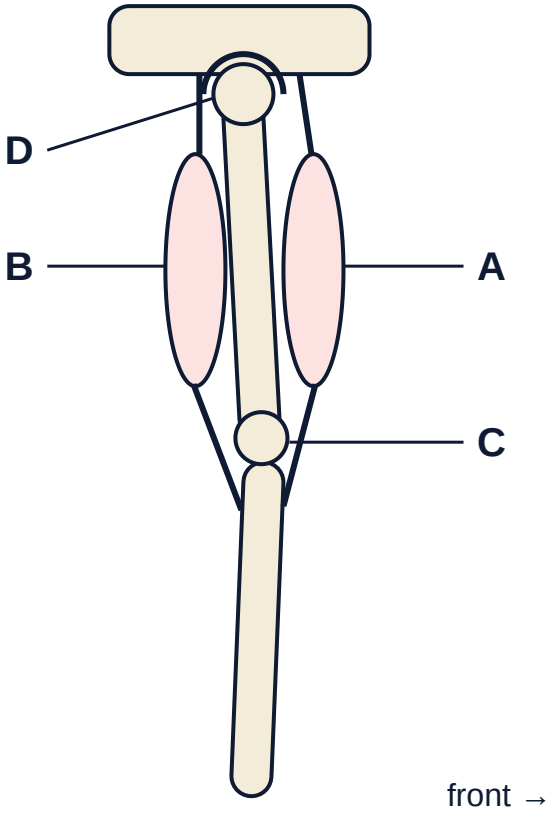
Answer all questions. Write your answers in the spaces provided. Show your working where there is space for it. The number of marks for each question is shown in brackets [ ].

Name _____ Class _____ Date _____ Mark: ____ / 100

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 45 minutes — 50 marks

1 On sports day, Salma takes part in the long jump. The diagram shows the bones and two muscles of her leg, seen from the side. The front of the leg is on the right. **[1]**



(a) Which letter, A, B, C or D, shows a hinge joint?

(b) Name the type of joint at D. **[1]**

(c) Salma straightens her leg at the knee to push off the ground. Which muscle, A or B, contracts to do this? **[1]**

(d) Muscles A and B are called an antagonistic pair. Explain what this means and why muscles must work in pairs. **[2]**

2 Youssef runs an 800 m race on sports day. His teacher counts his breathing rate before and after the race. **[1]**

Time	Breathing rate in breaths per minute
before the race	15
1 minute after the race	42
5 minutes after the race	24
10 minutes after the race	16

Use the table to answer the questions.

(a) By how many breaths per minute did Youssef's breathing rate increase from before the race to 1 minute after the race?

Working

Answer ____ breaths per minute

(b) Which describes what happens when Youssef breathes in? **[1]**

- ☐ A) The diaphragm relaxes and moves up; the ribs move down and in.
- ☐ B) The diaphragm contracts and moves down; the ribs move up and out.
- ☐ C) The diaphragm contracts and moves up; the ribs move down and in.
- ☐ D) The diaphragm relaxes and moves down; the ribs move up and out.

(c) Explain why Youssef's breathing rate is much higher just after the race. Use the word **respiration** in your answer. **[2]**

(d) Youssef says: "Breathing and respiration are the same thing." Explain why he is wrong. **[2]**

(e) The word equation for aerobic respiration is: **[1]**
glucose + oxygen → _____ + water
Write the missing product.

(f) Describe how oxygen gets from the air in the alveoli into Youssef's blood. **[2]**

(g) Which statement about aerobic respiration is correct? **[1]**

- ☐ A) It happens only in animal cells, inside the nucleus.
- ☐ B) It happens only in the lungs, when air is breathed in.
- ☐ C) It happens in the mitochondria of plant and animal cells and releases energy in a controlled way.
- ☐ D) It releases all the energy in glucose at once, like burning.

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3 Egypt has built a very large solar park in the desert near Aswan. Egypt also gets electricity from wind farms and from power stations that burn natural gas. **[1]**

(a) Which list contains **only** renewable energy resources?

- ☐ A) wind, solar, natural gas
- ☐ B) tidal, coal, wind
- ☐ C) solar, wind, tidal
- ☐ D) oil, natural gas, coal

(b) Suggest **one** reason why the desert in southern Egypt is a good place for a solar park. **[1]**

(c) A company now makes water bottles from bioplastic instead of plastic made from crude oil. Which is an advantage of bioplastic? **[1]**

- ☐ A) It is made from plants, which can be grown again, so it is renewable.
- ☐ B) It is made from coal, so it lasts for millions of years.
- ☐ C) It is made from crude oil, so it is cheaper.
- ☐ D) It can never break down, so it does not need recycling.

4 Mariam investigates how four metals, W, X, Y and Z, react with dilute hydrochloric acid. The metals are copper, iron, magnesium and zinc, in a random order. **[1]**
She adds 0.5 g of each metal powder to 20 cm^3 of the acid and measures the volume of gas made in 1 minute. She does each test three times.

Metal	Test 1 in cm^3	Test 2 in cm^3	Test 3 in cm^3
W	6	5	7
X	0	0	0
Y	52	50	51
Z	21	35	23

Use the results in the table to answer the questions.

(a) Write the letters of the metals in order of reactivity, starting with the **most** reactive.

(b) Which metal, W, X, Y or Z, is iron? **[1]**

(c) Which piece of equipment should Mariam use to measure the volume of gas? **[1]**

- ☐ A) a thermometer
- ☐ B) a top-pan balance
- ☐ C) a stopwatch
- ☐ D) a gas syringe

(d) One result in the table is anomalous. Write down this result. **[1]**

Working

Answer ____ cm^3

(e) Calculate the mean volume of gas for metal Z. Do not include the anomalous result. **[1]**

Working

Answer ____ cm^3

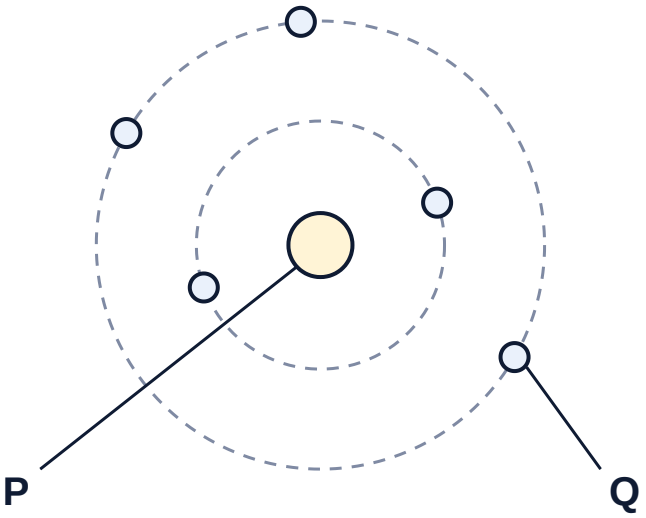
(f) Complete the word equation for the reaction of iron with hydrochloric acid. **[1]**

iron + hydrochloric acid → _____ + hydrogen

(g) Mariam's teacher puts a small piece of gold into the same acid. Predict what Mariam will see, and explain your prediction. **[2]**

(h) Dilute hydrochloric acid is an irritant. Describe **one** way Mariam can reduce the risk of harm. **[1]**

5 The diagram shows the Rutherford model of an atom. **[1]**



(a) Which letter, P or Q, shows a particle with a **negative** charge?

(b) Which row gives the correct charges of a proton and a neutron? **[1]**

Row	proton	neutron
A	positive	no charge
B	negative	positive
C	no charge	negative
D	positive	negative

Choose the correct row.

- ☐ A) A
- ☐ B) B
- ☐ C) C
- ☐ D) D

(c) An atom of fluorine has 9 electrons and 10 neutrons. How many protons does it have? **[1]**

Working

Answer

(d) Explain what holds the electrons in an atom. **[2]**

(e) Which statement describes the Rutherford model of the atom? **[1]**

☐ A) Positive charge is spread through a ball, with electrons dotted inside it like fruit in a cake.

☐ B) A tiny positive nucleus holds almost all the mass; electrons move around it, and most of the atom is empty space.

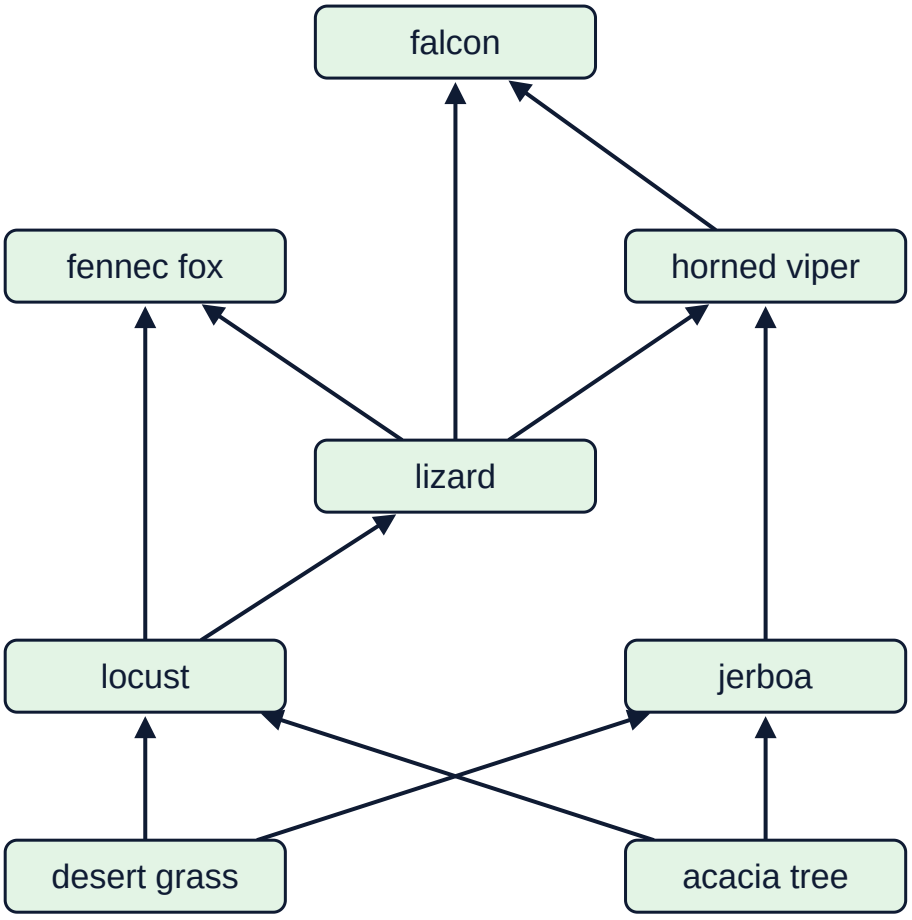
☐ C) An atom is a solid ball with no smaller particles inside it.

☐ D) A large negative nucleus has positive electrons moving around it.

6 The Western Desert of Egypt is an ecosystem. The diagram shows part of a desert food web. **[1]**
Farmers near an oasis spray a pesticide that is not broken down in living things. Scientists measure the pesticide in one food chain.

Organism	desert grass	locust	lizard	horned viper	falcon
Pesticide in parts per million	0.05	0.6	4.5	18	72

Use the food web and the table to answer the questions.



(a) Which of these is a **habitat** within the desert ecosystem?

- ☐ A) the hot, dry climate of the desert
- ☐ B) all the animals that live in the desert
- ☐ C) a small oasis pool with palm trees
- ☐ D) the food web in the diagram

(b) Name **one** producer in this food web. **[1]**

(c) Name the animal that competes with the lizard for locusts. **[1]**

(d) How many times greater is the concentration of pesticide in the falcon than in the locust? **[1]**

Working

Answer

(e) Explain why the concentration of pesticide is highest in the falcon. **[2]**

(f) Trucks bring a new species of rat to the oasis by accident. The rat breeds quickly and eats seeds and grass. Suggest **one** way the rat could make the number of jerboas fall. **[1]**

7 Athletes need a healthy diet and healthy blood and lungs. **[1]**

(a) Which row shows a good source of vitamin C and one of its functions?

- ☐ A) fish oil — helps the body take in calcium for strong bones
- ☐ B) carrots — needed for good eyesight in dim light
- ☐ C) red meat — needed to make haemoglobin in red blood cells
- ☐ D) oranges and guava — keep the skin and gums healthy

(b) The night before a long race, Karim eats a big plate of pasta. Pasta is rich in one nutrient that the body uses mainly as a source of energy. Name this nutrient. **[1]**

(c) Name the part of the blood that carries glucose from the small intestine to the muscles. **[1]**

(d) Explain why a person who smokes usually cannot run as far as a person who does not smoke. **[2]**

8 This question is about the Earth and space. **[1]**

(a) Which description of a galaxy is correct?

- ☐ A) a huge group of stars and planetary systems, with dust and gas, held together by gravity
- ☐ B) one star with the planets that orbit it
- ☐ C) a cloud of gas around one planet
- ☐ D) all the moons in the Solar System

(b) Which statement about asteroids is correct? **[1]**

- ☐ A) Asteroids are small stars that give out their own light.
- ☐ B) Asteroids are rocks, smaller than planets, left over from when the planetary system formed.
- ☐ C) Asteroids are clouds of gas between the planets.
- ☐ D) Asteroids are bigger than planets and orbit the Moon.

(c) Which statement describes the **climate** of Aswan, not its weather? **[1]**

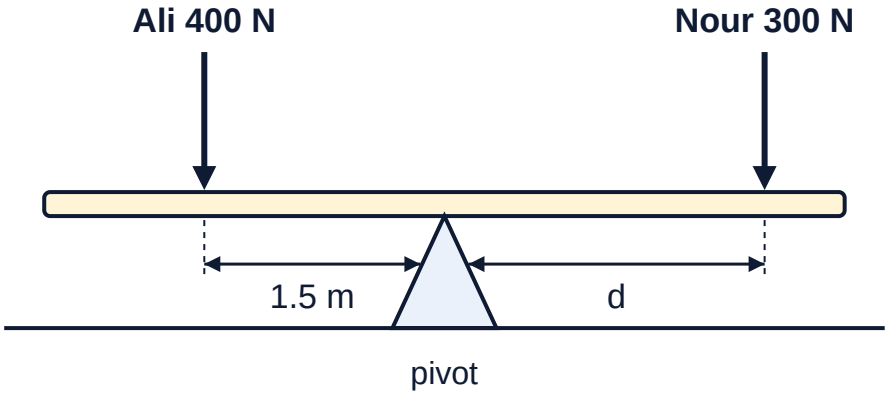
- ☐ A) It rained for ten minutes in Aswan yesterday.
- ☐ B) Tomorrow the temperature in Aswan will be 41 °C.
- ☐ C) There is a strong wind in Aswan this morning.
- ☐ D) Aswan has very hot, dry summers and almost no rain in most years.

(d) Scientists say that the Earth's climate has changed many times between ice ages and warm periods. Give **one** piece of evidence scientists use for this. **[1]**

(e) Name **one** gas in the atmosphere whose increase is making the Earth's climate warmer. **[1]**

Paper 2 — 45 minutes — 50 marks

1 Ali and Nour sit on a seesaw in the park. The seesaw is balanced. Ali weighs 400 N and sits 1.5 m from the pivot. Nour weighs 300 N and sits a distance d from the pivot. **[2]**



(a) Calculate the moment of Ali's weight about the pivot. Show your working.

Working

Answer _____ N m

(b) The seesaw is balanced. Calculate the distance d. Show your working. **[2]**

Working

Answer _____ m

(c) Nour moves and sits 1.5 m from the pivot instead. What happens? **[1]**

- ☐ A) Ali's end goes down, because Ali's moment is now bigger than Nour's.
- ☐ B) Nour's end goes down, because Nour's moment is now bigger than Ali's.
- ☐ C) The seesaw stays balanced, because they are both 1.5 m from the pivot.
- ☐ D) The seesaw stays balanced, because the pivot is in the middle.

(d) At home, Ali pushes on a door handle with a force of 15 N. The handle is 0.80 m from the hinges. Calculate the moment of Ali's push about the hinges. **[1]**

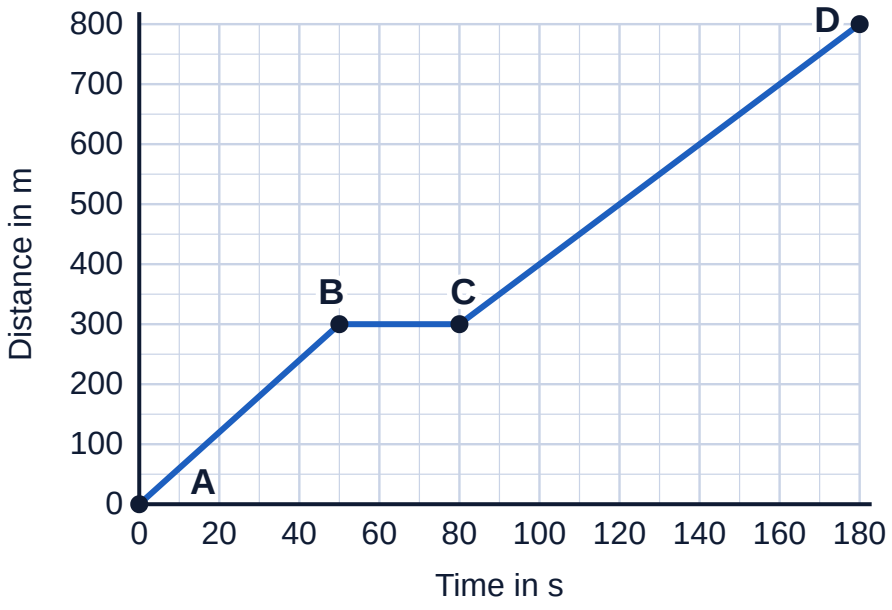
Working

Answer ____ N m

(e) Explain why door handles are fixed on the side of the door furthest from the hinges. **[2]**

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2 Karim runs in an 800 m race on sports day. The distance–time graph shows his race. He stops at B to tie his shoelace, then runs again from C. **[1]**



(a) For how many seconds does Karim stop?

Working

Answer ____ s

(b) Calculate Karim's speed between C and D. Show your working. **[2]**

Working

Answer ____ m/s

(c) Calculate Karim's average speed for the whole race. Give your answer to 1 decimal place. **[2]**

Working

Answer ____ m/s

(d) Between C and D Karim runs at a steady speed. Which statement about the forces on Karim is correct? **[1]**

- ☐ A) The forward force is bigger than the backward forces, so he speeds up.
- ☐ B) There are no forces acting on him.
- ☐ C) The forward force and the backward forces are balanced, so his speed does not change.
- ☐ D) The backward forces are bigger than the forward force, so he slows down.

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3 This question is about pressure and particles on sports day. **[2]**

(a) A shot-put athlete weighs 720 N. The total area of the soles of both her shoes touching the ground is 0.045 m². Calculate the pressure she puts on the ground. Show your working.

Working

Answer ____ Pa

(b) Sprinters wear shoes with small metal spikes underneath. Why do the spikes help them to grip the track? **[1]**

- ☐ A) The spikes increase the area touching the track, so the pressure is smaller.
- ☐ B) The spikes have a small area, so the pressure is large and they press into the track.
- ☐ C) The spikes make the runner's weight smaller.
- ☐ D) The spikes have a large area, so the pressure is large.

(c) A teacher pumps more air into a football. Use ideas about particles to explain why the pressure inside the ball increases. **[2]**

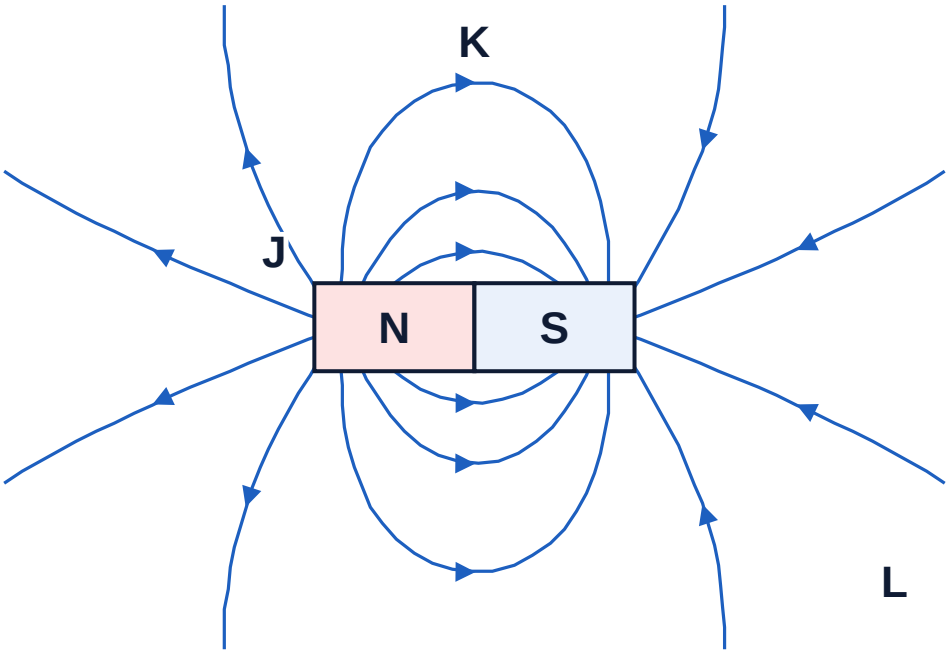
(d) Spectators at the far end of the field can smell the food being grilled. Use ideas about particles to explain how the smell reaches them. **[2]**

4 A scrapyard uses a crane with a large electromagnet to lift old cars. **[1]**

Sami makes a model of the crane. He winds 20 turns of insulated wire round an iron nail. He changes the current and counts how many small steel nails the electromagnet lifts.

Current in A	0.2	0.4	0.6	0.8	1.0
Number of steel nails lifted	3	6	9	5	15

Parts (e) to (j) are about Sami's investigation.



(a) The diagram shows the magnetic field around a bar magnet. At which point, J, K or L, is the magnetic field strongest?

(b) Which statement about the magnetic field lines is correct? **[1]**

- ☐ A) Outside the magnet they go from the S pole to the N pole.
- ☐ B) They are furthest apart near the poles.
- ☐ C) They cross each other near the middle of the magnet.
- ☐ D) Outside the magnet they go from the N pole to the S pole.

(c) Which of these is an electromagnet? **[1]**

- ☐ A) a steel nail stroked many times with a bar magnet
- ☐ B) a coil of insulated wire wrapped around an iron nail, with the wire connected to a cell
- ☐ C) an iron nail rubbed with a cloth
- ☐ D) a coil of insulated wire wrapped around an iron nail, with both ends of the wire not connected

(d) Explain why the scrapyard uses an electromagnet and not a large permanent magnet. **[1]**

(e) What is the independent variable in Sami's investigation? **[1]**

(f) One result is anomalous. At what current was this result? **[1]**

Working

Answer ____ A

(g) Predict how many nails the electromagnet would lift with a current of 1.2 A. **[1]**

Working

Answer ____ nails

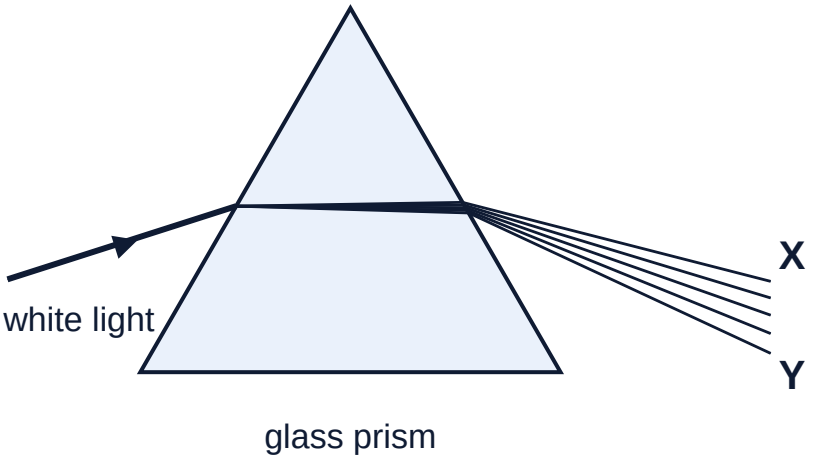
(h) Describe the pattern in Sami's results. Ignore the anomalous result. **[1]**

(i) Give **one** control variable Sami must keep the same. **[1]**

(j) Sami took each reading only once. Suggest how he could make his results more reliable, and explain why this helps. **[1]**

(k) The Earth also has a magnetic field. Complete the sentence with one word. **[1]**
The Earth has a magnetic field because its _____ acts like a magnet.

5 This question is about light and colour: a rainbow and the coloured lights on a school stage. **[1]**



(a) A rainbow forms when sunlight passes through raindrops. In the laboratory, Hana shines a ray of white light into a glass prism (diagram not drawn to scale). The white light splits into colours. What is the name for the splitting of white light into colours?

(b) Look at the diagram of Hana's prism. Which ray, X or Y, is red light? **[1]**

(c) Explain why the light changes direction when it goes from the air into the glass. **[1]**

(d) At a school concert, the stage is lit with coloured spotlights. A red spotlight and a green spotlight shine onto the same part of a white screen. What colour is seen where they overlap? **[1]**

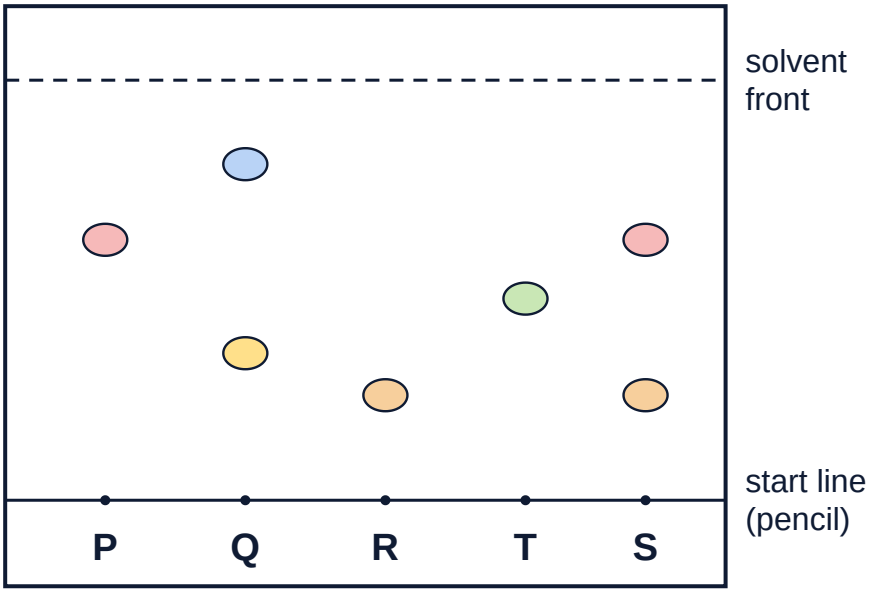
(e) On the stage, a singer wears a white shirt and blue trousers. Only a red spotlight shines on him. Describe and explain what colour his shirt and his trousers look. **[2]**

(f) A spotlight beam hits a flat mirror on the stage. The angle of incidence is 25°. What is the angle between the incident ray and the reflected ray? **[1]**

Working

Answer °

6 In the chemistry laboratory, Laila tests the colouring in an orange sports drink, S. She uses paper chromatography to compare it with four food dyes, P, Q, R and T. **[1]**



(a) Which **two** food dyes are in the sports drink S? Write two letters.

(b) Which food dye, P, Q, R or T, is a mixture of more than one substance? **[1]**

(c) Laila also makes some copper sulfate crystals. Her 20 g sample contains 18.4 g of copper sulfate. The rest is water and unreacted copper oxide. Calculate the percentage purity of Laila's copper sulfate. Show your working. **[2]**

Working

Answer ____ %

(d) Laila's copper sulfate is not pure. Suggest **one** reason why the product of a chemical reaction is often not pure. **[1]**

(e) Laila makes two sugary sports drinks. **[1]**

Drink	Mass of sugar in g	Volume of drink in cm^3
A	30	500
B	20	250

Which drink, A or B, is more concentrated?

(f) The school nurse keeps instant cold packs for sports injuries. When a cold pack is squeezed, a solid dissolves in the water inside. The temperature falls from 21 °C to 4 °C. Is dissolving this solid exothermic or endothermic? **[1]**

(g) The table shows the solubility of potassium nitrate at different temperatures. **[2]**

Temperature in °C	0	20	40	60	80
Solubility in g per 100 g of water	13	32	64	110	169

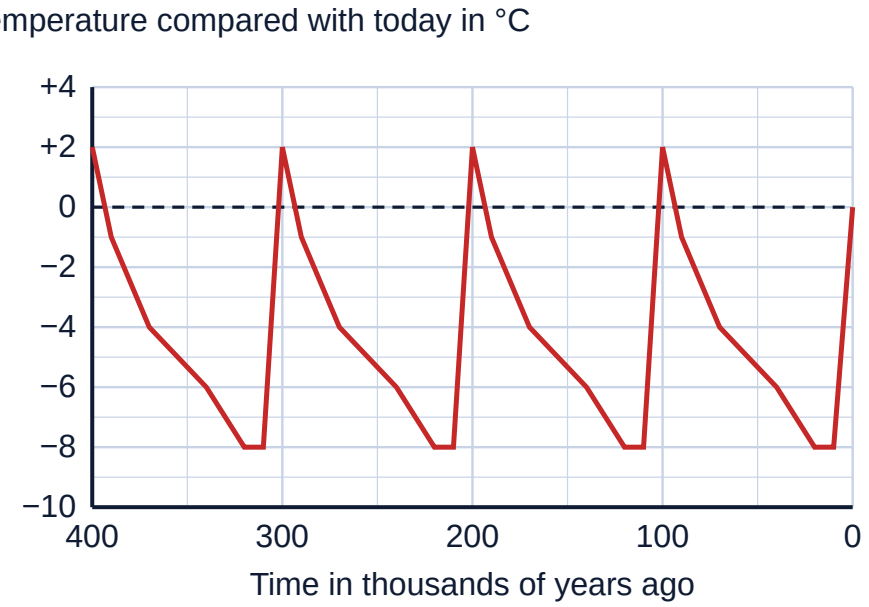
Laila makes a saturated solution of potassium nitrate in 100 g of water at 60 °C. She cools it to 20 °C. What mass of crystals forms? Show your working.

Working

Answer ____ g

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7 The graph shows how the average temperature of the Earth changed over the last 400 000 years, compared with the temperature today. The pattern is based on evidence from ice cores. **[1]**



(a) What is the lowest temperature shown on the graph, compared with today?

Working

Answer ____ °C

(b) The climate goes through a cycle of ice ages and warm periods. How many thousand years are there between one warm period and the next? **[1]**

Working

Answer ____ thousand years